AI for Food Security Education

GregHarvest

Group Name: Machine Not Learning

Group Members:

1. *CHAN HUAN YEE*
2. *DARREN HUANG HUAT YAO*
3. *JACKY BONG CHUN WUI*
4. *SENG ZHI JIE (Group Leader)*
5. *WILLIAM TAN JUN KHAI*

Table of Contents

Introduction 2
Background of Case Study 2
Chosen SDG 2
Problem Statement 3
Objectives 3
Review of Existing Solutions 4
Methodology 5
Results 6
Prototype Showcase system 7
AI Acknowledgement 12
Conclusion 12

Introduction

Background of Case Study

In ASEAN countries, small-scale farmers remain the backbone of food production. However, they face mounting challenges from unpredictable weather, rising costs, and limited access to modern technologies. At the same time, urban youngsters tend to lack comprehension of the reality of agriculture, considering food as a limitless commodity rather than the outcome of rigorous labour and resource consumption. This division adds to weak food systems and rising inequality between rural and urban residents.

AI offers a human-centered approach to modernizing agriculture through real-time advice on weather and farming techniques. By integrating AI into farming practices, farmers can receive real-time guidance on crop management, weather forecasting, and soil health. This could enable them to make informed decisions and improve yields. In education, AI-driven platforms empower students to engage with agricultural data, fostering innovation and building solutions that address food insecurity. Furthermore, AI-enabled financial tools can help farmers assess risks and access credit, supporting sustainable growth and resilience.

Chosen SDG

SDG 2: Zero Hunger

Using AI- driven solutions to tackle food insecurity and improving agricultural productivity.

SDG 4: Quality Education

Empowers students to innovate, equips farmers with digital literacy and fosters inclusive, lifelong learning that strengthens food security.

SDG 9: Industry, Innovation and Infrastructure

Promoting smart farming platforms and digital tools that modernize agriculture.

SDG 13: Climate Action

Using AI to help farmers adapt to climate change with real-time weather and farming advice.

Problem Statement

A significant digital and educational divide exists in ASEAN agriculture. Many small-scale farmers lack the digital literacy to adopt modern technologies, leaving them dependent on outdated practices. This results in inefficient farming methods such as overplanting where farmers grow more crops but earn less due to poor market alignment and supply chain inefficiencies.

Without access to real-time insights, farmers cannot anticipate or mitigate climate risks. When extreme weather strikes, they are left in vulnerable conditions, facing crop losses and financial instability. This cycle of inefficiency, risk, and limited knowledge prevents the region from building a sustainable, self-sufficient food system.

Additionally, farmers struggle with limited access to financial tools and market information. This will lead to some difficulties to secure fair prices or investments. This cycle consists of inefficiency, high risk, and economic vulnerability preventing the region from building a sustainable and self-sufficient food system.

Objectives

1. Provide farmers with AI-powered recommendations on the most suitable crops based on weather forecast, market demand to reduce supply chain disruptions. (SDG2: Zero Hunger)
2. Offer digital tools to monitor land use, track crop cycles, and manage agricultural assets for better productivity and sustainability. (SDG9: Industry, Innovation and Infrastructure)
3. Deliver predictive insights and real-time alerts on weather risks to help farmers take proactive measures. (SDG13: Climate Action)
4. Act as a virtual assistant that provides tailored advice and suggestions whenever necessary, ensuring farmers have continuous guidance. (SDG4: Quality Education)

Review of Existing Solutions

KissanAI and Handysense are two notable solutions in the agricultural technology space, but they approach farmer support differently. KissanAI is a multilingual, voice-enabled advisory platform that provides farmers with personalized guidance on crop selection, pest management, irrigation, and sustainability practices. Its strength lies in breaking literacy and language barriers, making advisory services more inclusive. However, it is primarily focused on advisory functions and depends heavily on connectivity, which can limit its reach among small-scale farmers in rural areas. On the other hand, Handysense is an IoT-based precision farming solution that uses smart sensors to monitor soil moisture, temperature, and humidity, delivering real-time insights for automated irrigation and crop management. This makes it strong in hardware integration and resource optimization. Unfortunately, its reliance on sensor hardware can be costly and technically challenging for small-scale farmers.

Despite their strengths, both solutions leave gaps. KissanAI does not fully address farm management or financial empowerment, while Handysense lacks advisory and market guidance. Neither provides integrated financial tools to help farmers secure fair prices or investments, nor do they offer localized insights tailored to specific regions. The project aims to bridge these gaps by combining advisory support, real-time monitoring, land and asset management, climate risk prevention, and financial guidance into one accessible platform designed specifically for small-scale farmers. Therefore, this holistic approach positions our solution as more comprehensive and impactful compared to existing competitors.

Methodology

For this prototype, lightweight web application is developed using HTML, CSS and JavaScript to enable rapid development within the limited hackathon timeframe. The application is served using a Flask API, with the core back-end logic implemented in Python.

Before implementation, the overall workflow of the application was discussed and mapped out. Our front-end engineer then created a Figma prototype to design the user interface and visualize the user experience. Based on this prototype, the webpages were implemented and rendered using Flask's Jinja templating engine.

On the back-end, OpenWeather API (basic tier) is integrated to retrieve real-time weather data for a selected city, which is displayed on the main interface. In addition, OpenAI API is incorporated to power a chatbot assistant. By passing the retrieved weather data as context, the chatbot can generate personalized suggestions for users based on current weather conditions.

Through the integration of real-time environmental data and AI-driven analysis, the system demonstrates how simple technologies can be combined to deliver more context-aware and practical decision support for users.

Results

Our project outcomes largely met the expectations that had established at the beginning of development. Although several planned ideas could not be fully implemented due to time constraints, these concepts were still presented in the prototype and demonstrated during the project showcase. In addition, the current version of the application was intentionally designed with flexibility for future enhancements.

The application is primarily targeted at farmers in Southeast Asia, particularly those who have limited familiarity with artificial intelligence technologies. For this reason, the interface and overall user experience were designed to be simple, accessible, and easy to understand. The objective is to ensure that farmers, regardless of educational background, are able to use the application effectively as a practical advisory tool for planting decisions.

The developed application provides key environmental information, including weather conditions, humidity, wind speed, and dew point. These features assist farmers in assessing daily field conditions and making more informed decisions regarding irrigation and crop management. In addition, the Greg AI feature allows users to ask questions and upload documents to obtain recommendations on suitable crops and other agriculture-related concerns. A voice input function was also incorporated to improve accessibility and ease of use, particularly for users who may find text-based interaction less convenient.

Prototype Showcase system

Figure 1.1 shows the login page of the GregHarvest system. Users are allowed to login with Google, Apple, Phone or "Continue as Guest" for quick access.

Figure 1.1: Login Page of GregHarvest System

Figure 1.2 shows the pre-setup page of GregHarvest System. This page allows users to input details of the location, land size and crops with its market price. Ideally the market price will be generated by our AI based on the current market price.

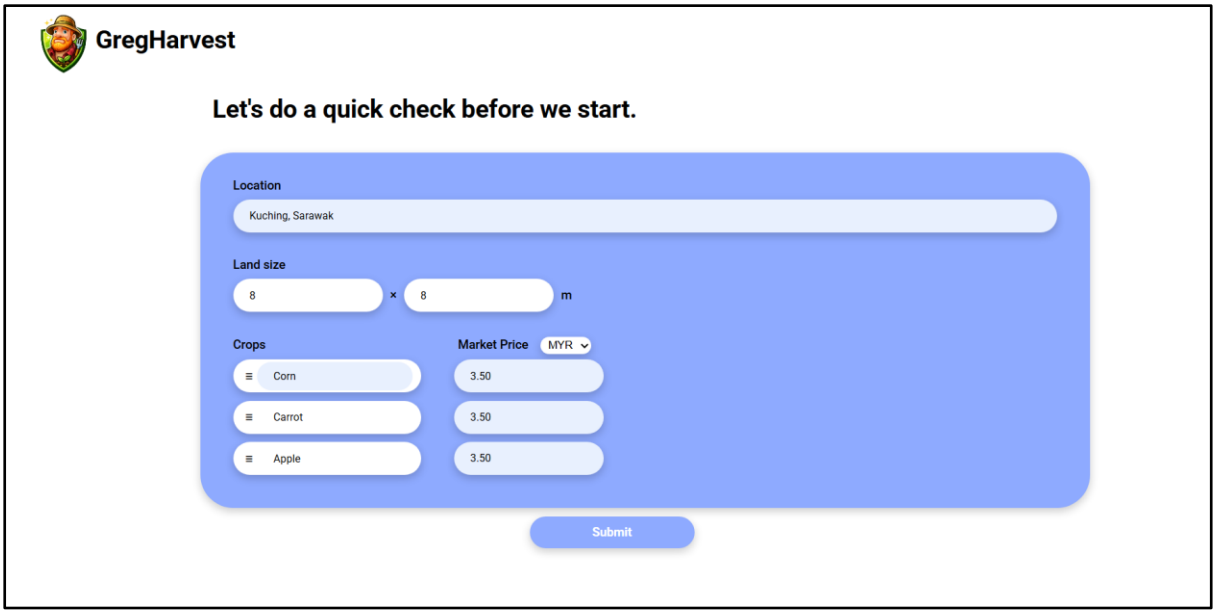

Figure 1.2: Pre-setup page of GregHarvest System

Figure 1.3 shows the profile page of GregHarvest System after users submitting the pre-setup page. The main feature of this page is the grid view, which is generated by our AI to plan the crop distribution. User can then adjust the distribution according to their preference.

Figure 1.3: Profile page of GregHarvest System

Figure 1.4 shows the main page of our system. At the top right of the page, there are three buttons with different functions. The center of the page displays the weather card, which provides all the necessary information. Additionally, there is a chat input button that allows users to ask questions to our AI.

Figure 1.4: Main Page of GregHarvest System

Figure 1.5 illustrates the first button, located at the top right that serves as a crop management quick access feature. This button provides users with direct entry to crop-related functions, enabling them to efficiently monitor, adjust and organise their crops distribution.

Figure 1.5: GregHarvest Crop Management Quick Access Feature

Figure 1.6 shows the interface with a 'More' button that allows users to navigate between 'Home', 'Ask AI', 'Profile', Chat history records and 'Log Out'. The button beside it provides quick access to the profile management page that enables user to manage their personal settings and information efficiently.

Figure 1.6: The More and Profile Buttons in GregHarvest System

Figure 1.7 shows the chat input feature that allows users to ask farm-related questions. User can interact with the AI in our system by typing text queries, uploading pictures or files, and even using the available speech-to-text function.

Figure 1.7: Chat Input Feature for Farm Queries

Figure 1.8 shows the chatbot conversation page where the questions asked by users are redirected for interaction with the AI. This page can also be accessed through the 'Ask AI' option in the 'More' menu. The focus of this page is to facilitate direct conversations between users and the AI that enables personalised guidance and support.

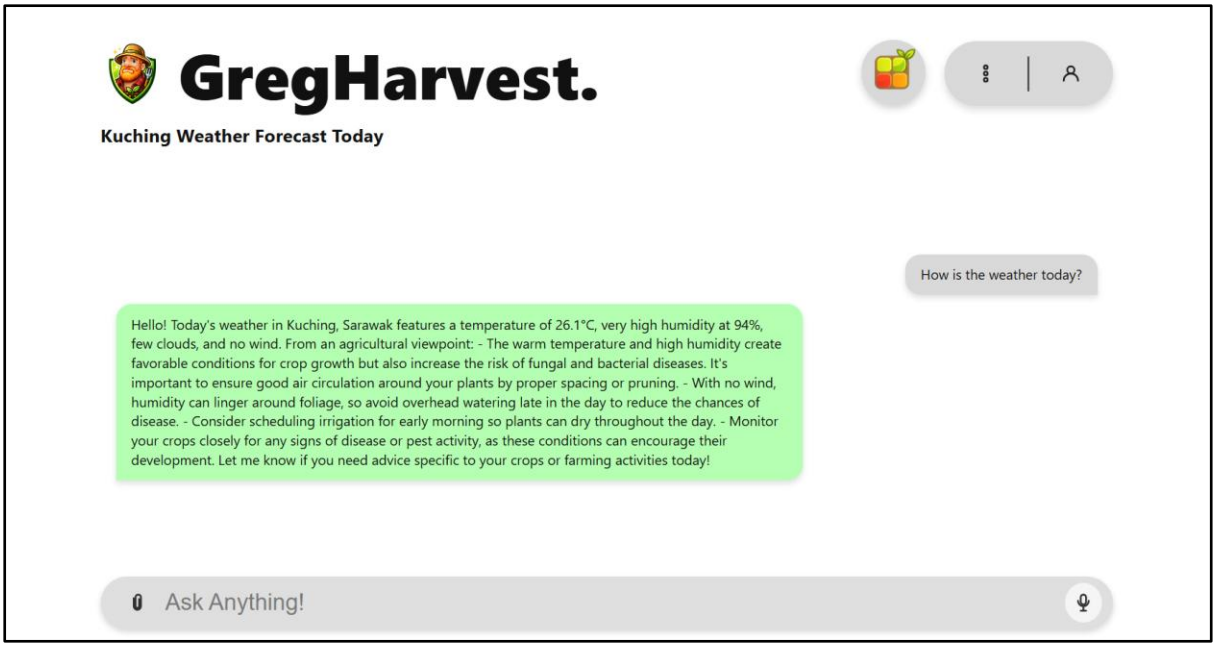

Figure 1.8: GregHarvest Chatbot Conversation Page

In Figure 1.9, the chatbot illustrates the memory-based conversation feature. The AI utilises information such as location, weather conditions, and crop management data to deliver more accurate analysis and tailored advice. This ensures that recommendations are context-aware and personalised to the user's farming activities.

Figure 1.9: Memory-Based Chatbot Conversation

AI Acknowledgement

The AI tools used in this project were ChatGPT, Codex, and Microsoft Copilot. ChatGPT and Codex were utilized primarily for code review and improvement, including checking logic, identifying possible errors, and suggesting refinements. GitHub Copilot was also used to support our learning process by providing coding suggestions and helping us deepen our understanding of programming practices. ChatGPT also used to generate our brand logo.

Conclusion

This project addresses the critical challenges faced by small-scale farmers in ASEAN agriculture, including the digital divide, inefficient farming practices, vulnerability to climate risks, and limited access to financial tools and market information. By integrating AI-powered crop recommendations, digital land and asset management, predictive climate insights, and on-demand advisory support, the system empowers farmers to make informed decisions, improve productivity and strengthen resilience.

In terms of future development, several improvements have been identified. These include the introduction of a premium version with more advanced capabilities, such as sensor integration and real-time field monitoring. Another important enhancement is the implementation of multilingual support, enabling farmers from different countries in Southeast Asia to access the application in their native languages. Overall, the results indicate that the application has strong potential to serve as a practical and relevant tool for farmers across the Southeast Asian region.